

deportation (Méndez et al., 2020). While we identified an increased risk of diabetes EDVs among people who are Hispanic, there are likely additional health inequities that we were unable to detect in our study.

5. Conclusion

We found that smoke events were associated with increased risk of EDVs for all respiratory diseases combined, asthma, CLRD, all cardiovascular diseases combined, and schizophrenia and decreased risk of EDVs for substance use disorders and suicide. There was some indication of effect modification by race/ethnicity, including a higher risk of EDVs for diabetes among people who are Hispanic. With the growing risk of more frequent and more severe wildfires due to climate change, understanding the differential health effects of wildfire smoke and identifying the structural barriers faced by the most susceptible populations is critical. These findings may help inform public health practice to protect populations that are disproportionately impacted by wildfire smoke and to plan for adequate healthcare services during and after smoke events.

Funding sources

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Institutional Review Board statement

Institutional Review Board approval was obtained from California's Committee for the Protection of Human Subjects (project number 12-04-0075).

CRediT authorship contribution statement

Annie I. Chen: Conceptualization, Methodology, Data curation, Investigation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing. **Keita Ebisu:** Methodology, Data curation, Formal analysis, Visualization, Writing – review & editing. **Tarik Benmarhnia:** Methodology, Data curation, Writing – review & editing. **Rupa Basu:** Conceptualization, Methodology, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The health data that were used are confidential. Daily, ZCTA-level wildfire-specific PM_{2.5} concentrations are available here: https://github.com/benmarhnia-lab/Wildfire_PM25_California_ZIP. Air quality and meteorological data from monitors are publicly available as cited in the text. Data from the 2015–2019 5-year American Community Survey may be obtained from the Census Bureau via the tidycensus package in R.

Acknowledgements

We thank Dharshani Pearson, MPH for preparing the original temperature dataset. We would like to thank Xiangmei Wu, Ph.D., Amal Syed, MPH, and Susan Hurley, MPH for their review and valuable feedback. The opinions expressed in this article are those of the authors and do not represent those of the California Environmental Protection Agency or the Office of Environmental Health Hazard Assessment.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envres.2023.117154>.

References

- Aguilera, R., Corringham, T., Gershunov, A., Leibel, S., Benmarhnia, T., 2021a. Fine particles in wildfire smoke and pediatric respiratory health in California. *Pediatrics* 147, e2020027128. <https://doi.org/10.1542/peds.2020-027128>.
- Aguilera, R., Corringham, T.W., Gershunov, A., Benmarhnia, T., 2021b. Wildfire smoke impacts respiratory health more than fine particles from other sources: observational evidence from southern California. *Nat. Commun.* 12, 1493. <https://doi.org/10.1038/s41467-021-21708-0>.
- Aguilera, R., Luo, N., Basu, R., Wu, J., Clemesha, R., Gershunov, A., Benmarhnia, T., 2023. A novel ensemble-based statistical approach to estimate daily wildfire-specific PM_{2.5} in California (2006–2020). *Environ. Int.* 171, 107719. <https://doi.org/10.1016/j.envint.2022.107719>.
- Altman, D.G., 2003. Statistics Notes: interaction revisited: the difference between two estimates. *BMJ* 326. <https://doi.org/10.1136/bmj.326.7382.219>, 219–219.
- Amjad, S., Chojecki, D., Osornio-Vargas, A., Ospina, M.B., 2021. Wildfire exposure during pregnancy and the risk of adverse birth outcomes: a systematic review. *Environ. Int.* 156, 106644. <https://doi.org/10.1016/j.envint.2021.106644>.
- Basu, R., Feng, W.-Y., Ostro, B.D., 2008. Characterizing temperature and mortality in nine California counties. *Epidemiology* 19, 138. <https://doi.org/10.1097/EDE.0b013e31815c1da7>.
- Basu, R., Gavin, L., Pearson, D., Ebisu, K., Malig, B., 2018. Examining the association between apparent temperature and mental health-related emergency room visits in California. *Am. J. Epidemiol.* 187, 726–735. <https://doi.org/10.1093/aje/kwx295>.
- Basu, R., Pearson, D., Malig, B., Broadwin, R., Green, R., 2012. The effect of high ambient temperature on emergency room visits. *Epidemiology* 23, 813–820. <https://doi.org/10.1097/EDE.0b013e31826b7f97>.
- Black, C., Gerriets, J.E., Fontaine, J.H., Harper, R.W., Kenyon, N.J., Tablin, F., Schelegle, E.S., Miller, L.A., 2017. Early life wildfire smoke exposure is associated with immune dysregulation and lung function decrements in adolescence. *Am. J. Respir. Cell Mol. Biol.* 56, 657–666. <https://doi.org/10.1165/rcmb.2016-0380OC>.
- Block, M.L., Calderón-Garcidueñas, L., 2009. Air pollution: mechanisms of neuroinflammation and CNS disease. *Trends Neurosci.* 32, 506–516. <https://doi.org/10.1016/j.tins.2009.05.009>.
- Boaggio, K., LeDuc, S.D., Rice, R.B., Duffney, P.F., Foley, K.M., Holder, A., McDow, S., Weaver, C.P., 2022. Beyond particulate matter mass: heightened levels of lead and other pollutants associated with destructive fire events in California. *Environ. Sci. Technol.* <https://doi.org/10.1021/acs.est.2c02099>.
- Bretler, T., Weisberg, H., Koren, O., Neuman, H., 2019. The effects of antipsychotic medications on microbiome and weight gain in children and adolescents. *BMC Med.* 17, 112. <https://doi.org/10.1186/s12916-019-1346-1>.
- Brown, M.R.G., Pazderka, H., Agyapong, V.I.O., Greenshaw, A.J., Cribben, I., Brett-MacLean, P., Drolet, J., McDonald-Harker, C.B., Omeje, J., Lee, B., Mankowski, M., Noble, S., Kitching, D.T., Silverstone, P.H., 2021. Mental health symptoms unexpectedly increased in students aged 11–19 Years during the 3.5 Years after the 2016 fort McMurray wildfire: findings from 9,376 survey responses. *Front. Psychiatry* 12.
- Budiman, A., Ruiz, N.G., 2021a. Key Facts about Asian Americans, a Diverse and Growing Population. Pew Research Center. URL: <https://www.pewresearch.org/fact-tank/2021/04/29/key-facts-about-asian-americans/>. (Accessed 3 March 2023).
- Budiman, A., Ruiz, N.G., 2021b. Key Facts about Asian Origin Groups in the U.S. Pew Research Center. URL: <https://www.pewresearch.org/fact-tank/2021/04/29/key-facts-about-asian-origin-groups-in-the-u-s/>. (Accessed 3 March 2023).
- CAL FIRE, 2022. Statistics [WWW Document]. URL: <https://www.fire.ca.gov/our-impact/statistics>. accessed 2.1.23.
- California Air Resources Board, 2021. Camp Fire Air Quality Data Analysis. California Air Resources Board.
- California Department of Forestry and Fire Protection, 2022. GIS Data [WWW Document]. URL: <https://frap.fire.ca.gov/mapping/gis-data/>. accessed 2.1.23.
- California Department of Health Care Access and Information (HCAI), 2021. Data Documentation [WWW Document]. HCAI. URL: <https://hcai.ca.gov/data-and-reports/request-data/data-documentation/>. accessed 2.1.22.
- Carberry, C.K., Koval, L.E., Payton, A., Hartwell, H., Ho Kim, Y., Smith, G.J., Reif, D.M., Jaspers, I., Jan Gilmour, M., Rager, J.E., 2022. Wildfires and extracellular vesicles: exosomal MicroRNAs as mediators of cross-tissue cardiopulmonary responses to biomass smoke. *Environ. Int.* 167, 107419. <https://doi.org/10.1016/j.envint.2022.107419>.
- Cascio, W.E., Cascio, Wayne E., Cascio, Wayne E., 2018. Wildland fire smoke and human health. *Sci. Total Environ.* 624, 586–595. <https://doi.org/10.1016/j.scitotenv.2017.12.086>.
- Chen, H., Samet, J.M., Bromberg, P.A., Tong, H., 2021. Cardiovascular health impacts of wildfire smoke exposure. Part. Fibre Toxicol. 18. <https://doi.org/10.1186/s12989-020-00394-8>, 2–2.
- Clinical Classifications Software Refined (CCSR) for ICD-10-CM Diagnoses, 2022 [WWW Document]. URL: <https://hcup-us.ahrq.gov/toolssoftware/ccsr/dxcsr.jsp>. (Accessed 2 August 2023).
- Comer, A.L., Carrier, M., Tremblay, M.-È., Cruz-Martín, A., 2020. The inflamed brain in schizophrenia: the convergence of genetic and environmental risk factors that lead to

- uncontrolled neuroinflammation. *Front. Cell. Neurosci.* 14, 274. <https://doi.org/10.3389/fncel.2020.00274>.
- Delfino, R.J., Brummel, S.S., Wu, J., Stern, H.S., Ostro, B., Lipsett, M., Lipsett, M., Winer, A.M., Street, D.H., Zhang, L., Tjoa, T., Gillen, D.L., 2009. The relationship of respiratory and cardiovascular hospital admissions to the southern California wildfires of 2003. *Occup. Environ. Med.* 66, 189–197. <https://doi.org/10.1136/oem.2008.041376>.
- Di, Q., Amini, H., Shi, L., Kloog, I., Silvern, R., Kelly, J., Sabath, M.B., Choirat, C., Koutrakis, P., Lyapustin, A., Wang, Y., Mickley, L.J., Schwartz, J., 2019. An ensemble-based model of PM_{2.5} concentration across the contiguous United States with high spatiotemporal resolution. *Environ. Int.* 130, 104909 <https://doi.org/10.1016/j.envint.2019.104909>.
- Eisenman, D.P., Galway, L.P., 2022. The mental health and well-being effects of wildfire smoke: a scoping review. *BMC Publ. Health* 22, 2274. <https://doi.org/10.1186/s12889-022-14662-z>.
- Eisenman, D.P., Kyaw, M.M.T., Eclarino, K., 2021. Review of the Mental Health Effects of Wildfire Smoke, Solastalgia, and Non-traditional Firefighters. UCLA Center for Healthy Climate Solutions, David Geffen School of Medicine at UCLA, and Climate Resolve.
- Elser, H., Rowland, S.T., Marek, M.S., Kiang, M.V., Shea, B., Do, V., Benmarhnia, T., Schneider, A.L.C., Casey, J.A., 2023. Wildfire smoke exposure and emergency department visits for headache: a case-crossover analysis in California, 2006–2020. *Headache* 63, 94–103. <https://doi.org/10.1111/head.14442>.
- Feo, T.J., Mace, A.J., Brady, S.E., Lindsey, B., 2020. The Costs of Wildfire in California. California Council on Science and Technology.
- Gao, Q., Xu, Q., Guo, X., Fan, H., Zhu, H., 2017. Particulate matter air pollution associated with hospital admissions for mental disorders: a time-series study in Beijing, China. *Eur. Psychiatr.* 44, 68–75. <https://doi.org/10.1016/j.eurpsy.2017.02.492>.
- Giussani, D.A., Niu, Y., Herrera, E.A., Richter, H.G., Camm, E.J., Thakor, A.S., Kane, A.D., Hansell, J.A., Brain, K.L., Skeffington, K.L., Itani, N., Wooding, F.B.P., Cross, C.M., Allison, B.J., 2014. Heart disease link to fetal hypoxia and oxidative stress. In: Zhang, L., Ducas, C.A. (Eds.), *Advances in Fetal and Neonatal Physiology*, Advances in Experimental Medicine and Biology. Springer, New York, NY, pp. 77–87. https://doi.org/10.1007/978-1-4939-1031-1_7.
- Haddad, P.M., Brain, C., Scott, J., 2014. Nonadherence with antipsychotic medication in schizophrenia: challenges and management strategies. *Patient Relat. Outcome Meas.* 5, 43–62. <https://doi.org/10.2147/PROM.S42735>.
- Heaney, A., Stowell, J.D., Liu, J.C., Basu, R., Marlier, M., Kinney, P., 2022. Impacts of fine particulate matter from wildfire smoke on respiratory and cardiovascular health in California. *GeoHealth* 6, e2021GH000578. <https://doi.org/10.1029/2021GH000578>.
- Hoshiko, S., Buckman, J.R., Jones, C.G., Yeomans, K.R., Mello, A., Thilakarathne, R., Sergienko, E., Allen, K., Bello, L., Rappold, A.G., 2023. Responses to wildfire and prescribed fire smoke: a survey of a medically vulnerable adult population in the wildland-urban interface, Mariposa county, California. *Int. J. Environ. Res. Publ. Health* 20, 1210. <https://doi.org/10.3390/ijerph20021210>.
- Isaac, F., Toukhsati, S.R., Klein, B., DiBenedetto, M., Kennedy, G.A., 2023. Prevalence and predictors of sleep and trauma symptoms in wildfire survivors. *Sleep Epidemiology* 3, 100052. <https://doi.org/10.1016/j.sleepe.2022.100052>.
- Jennings, V., Johnson Gaither, C., 2015. Approaching environmental health disparities and green spaces: an ecosystem services perspective. *Int. J. Environ. Res. Publ. Health* 12, 1952–1968. <https://doi.org/10.3390/ijerph120201952>.
- Kim, Y.H., Tong, H., Daniels, M., Boykin, E., Krantz, Q.T., McGee, J., Hays, M., Kovalick, K., Dye, J.A., Gilmour, M.I., 2014. Cardiopulmonary toxicity of peat wildfire particulate matter and the predictive utility of precision cut lung slices. *Part. Fibre Toxicol.* 11, 29. <https://doi.org/10.1186/1743-8977-11-29>.
- Kim, Y.H., Warren, S.H., Krantz, Q.T., King, C., Jaskot, R., Preston, W.T., George, B.J., Hays, M.D., Landis, M.S., Higuchi, M., DeMarini, D.M., Gilmour, M.I., 2018. Mutagenicity and lung toxicity of smoldering vs. flaming emissions from various biomass fuels: implications for health effects from wildland fires. *Environ. Health Perspect.* 126, 017011 <https://doi.org/10.1289/EHP2200>.
- Kondo, M.C., De Roos, A.J., De Roos, A.J., White, L.S., Heilman, Warren E., Heilman, W. E., Mockrin, Miranda H., Mockrin, M.H., Gross-Davis, C.A., Burstyn, I., 2019. Meta-analysis of heterogeneity in the effects of wildfire smoke exposure on respiratory health in North America. *Int. J. Environ. Res. Publ. Health* 16, 960. <https://doi.org/10.3390/ijerph16060960>.
- Kondo, M.C., Reid, C.E., Mockrin, M.H., Heilman, W.E., Long, D.D., 2022. Socio-demographic and health vulnerability in prescribed-burn exposed versus unexposed counties near the National Forest System. *Sci. Total Environ.* 806, 150564 <https://doi.org/10.1016/j.scitotenv.2021.150564>.
- Kroken, R.A., Sommer, I.E., Steen, V.M., Dieset, I., Johnsen, E., 2018. Constructing the immune signature of schizophrenia for clinical use and research; an integrative review translating descriptives into diagnostics. *Front. Psychiatr.* 9, 753. <https://doi.org/10.3389/fpsy.2018.00753>.
- Lane, H.M., Morello-Frosch, R., Marshall, J.D., Apte, J.S., 2022. Historical redlining is associated with present-day air pollution disparities in U.S. Cities. *Environ. Sci. Technol. Lett.* 9, 345–350. <https://doi.org/10.1021/acs.estlett.1c01012>.
- Liu, J.C., Peng, R.D., 2019. The impact of wildfire smoke on compositions of fine particulate matter by ecoregion in the Western US. *J. Expo. Sci. Environ. Epidemiol.* 29, 765–776. <https://doi.org/10.1038/s41370-018-0064-7>.
- Liu, J.C., Pereira, G., Uhl, S.A., Bravo, Mercedes A., Bravo, M.A., Bell, M.L., 2015. A systematic review of the physical health impacts from non-occupational exposure to wildfire smoke. *Environ. Res.* 136, 120–132. <https://doi.org/10.1016/j.envres.2014.10.015>.
- Liu, J.C., Wilson, A., Mickley, L.J., Dominici, F., Ebisu, K., Wang, Y., Wang, Y., Wang, Y., Sulprizio, M.P., Peng, R.D., Yue, X., Son, J.-Y., Anderson, G.E., Anderson, G.B., Bell, M.L., 2017a. Wildfire-specific fine particulate matter and risk of hospital admissions in urban and rural counties. *Epidemiology* 28, 77–85. <https://doi.org/10.1097/ede.0000000000000556>.
- Liu, J.C., Wilson, A., Mickley, L.J., Ebisu, K., Sulprizio, M.P., Wang, Y., Peng, R.D., Yue, X., Dominici, F., Bell, M.L., 2017b. Who among the elderly is most vulnerable to exposure to and health risks of fine particulate matter from wildfire smoke? *Am. J. Epidemiol.* 186, 730–735. <https://doi.org/10.1093/aje/kwx141>.
- Makkonen, U., Hellén, H., Anttila, P., Ferm, M., 2010. Size distribution and chemical composition of airborne particles in south-eastern Finland during different seasons and wildfire episodes in 2006. *Sci. Total Environ.* 408, 644–651. <https://doi.org/10.1016/j.scitotenv.2009.10.050>.
- Malig, B., Fairley, D., Fairley, David, Pearson, D., Wu, X.M., Ebisu, K., Basu, R., 2021. Examining fine particulate matter and cause-specific morbidity during the 2017 North San Francisco Bay wildfires. *Sci. Total Environ.* 787, 147507 <https://doi.org/10.1016/j.scitotenv.2021.147507>.
- McCann, T.V., Clark, E., Lu, S., 2009. Subjective side effects of antipsychotics and medication adherence in people with schizophrenia. *J. Adv. Nurs.* 65, 534–543. <https://doi.org/10.1111/j.1365-2648.2008.04906.x>.
- Méndez, M., Flores-Haro, G., Zucker, L., 2020. The (in)visible victims of disaster: understanding the vulnerability of undocumented Latino/a and indigenous immigrants. *Geoforum* 116, 50–62. <https://doi.org/10.1016/j.geoforum.2020.07.007>.
- Moore, D., Copes, R., Fisk, R., Joy, R., Chan, K., Brauer, M., 2006. Population health effects of air quality changes due to forest fires in British Columbia in 2003. *Can. J. Public Health* 97, 105–108. <https://doi.org/10.1007/BF03405325>.
- Müller, N., Schwarz, M.J., 2010. Immune system and schizophrenia. *Curr. Immunol. Rev.* 6, 213–220. <https://doi.org/10.2174/157339510791823673>.
- National Oceanic and Atmospheric Administration, 2021. National Centers for environmental information (NCEI) [WWW Document]. URL: <https://www.noaa.gov/>. accessed 9.7.21.
- Newbury, J.B., Arseneault, L., Beevers, S., Kitwiroon, N., Roberts, S., Pariante, C.M., Kelly, F.J., Fisher, H.L., 2019. Association of air pollution exposure with psychotic experiences during adolescence. *JAMA Psychiatr.* 76, 614–623. <https://doi.org/10.1001/jamapsychiatry.2019.0056>.
- Nguyen, A.-M., Malig, B.J., Basu, R., 2021. The association between ozone and fine particles and mental health-related emergency department visits in California, 2005–2013. *PLoS One* 16, e0249675. <https://doi.org/10.1371/journal.pone.0249675>.
- Office of Water Use Efficiency, California Department of Water Resources, 2020. CIMIS: California irrigation management system [WWW Document]. URL: <https://cimis.water.ca.gov/>.
- R Core Team, 2021. R: A Language and Environment for Statistical Computing.
- Rappold, A.G., Reyes, J., Pouliot, G., Cascio, W.E., Diaz-Sanchez, D., Diaz-Sanchez, D., 2017. Community vulnerability to health impacts of wildland fire smoke exposure. *Environmental Science & Technology* 51, 6674–6682. <https://doi.org/10.1021/acs.est.6b06200>.
- Reid, C.E., Brauer, M., Johnston, F.H., Jerrett, M., Jerrett, Michael, Balmes, J.R., Elliott, C.T., 2016a. Critical review of health impacts of wildfire smoke exposure. *Environmental Health Perspectives* 124, 1334–1343. <https://doi.org/10.1289/ehp.1409277>.
- Reid, C.E., Considine, E.M., Maestas, M.M., Li, G., 2021. Daily PM_{2.5} concentration estimates by county, ZIP code, and census tract in 11 western states 2008–2018. *Sci. Data* 8. <https://doi.org/10.1038/s41597-021-00891-1>.
- Reid, C.E., Jerrett, M., Tager, I.B., Petersen, M.L., Mann, J.K., Balmes, J.R., 2016b. Differential respiratory health effects from the 2008 northern California wildfires: a spatiotemporal approach. *Environ. Res.* 150, 227–235. <https://doi.org/10.1016/j.envres.2016.06.012>.
- Robinson, T., Robertson, N., Curtis, F., Darko, N., Jones, C.R., 2023. Examining psychosocial and economic barriers to green space access for racialised individuals and families: a narrative literature review of the evidence to date. *Int. J. Environ. Res. Publ. Health* 20, 745. <https://doi.org/10.3390/ijerph20010745>.
- RStudio Team, 2020. RStudio. Integrated Development for R.
- Schranz, C.I., Castillo, E.M., Vilke, G.M., 2010. The 2007 San Diego wildfire impact on the emergency Department of the University of California, San Diego hospital system. *Prehospital Disaster Med.* 25, 472–476. <https://doi.org/10.1017/S1049023X0000858X>.
- Scieszka, D., Hunter, R., Begay, J., Bitsui, M., Lin, Y., Galewsky, J., Morishita, M., Klaver, Z., Wagner, J., Harkema, J.R., Herbert, G., Lucas, S., McVeigh, C., Bolt, A., Bleske, B., Canal, C.G., Mostovenko, E., Ottens, A.K., Gu, H., Campen, M.J., Noor, S., 2022. Neuroinflammatory and neurometabolic consequences from inhaled wildfire smoke-derived particulate matter in the western United States. *Toxicol. Sci.* 186, 149–162. <https://doi.org/10.1093/toxsci/kfab147>.
- Sera, F., Armstrong, B., Blangiardo, M., Gasparri, A., 2019. An extended mixed-effects framework for meta-analysis. *Stat. Med.* 38, 5429–5444. <https://doi.org/10.1002/sim.8362>.
- Silveira, S., Kornbluh, M., Withers, M.C., Grennan, G., Ramanathan, V., Mishra, J., 2021. Chronic mental health sequelae of climate change extremes: a case study of the deadliest Californian wildfire. *Int. J. Environ. Res. Publ. Health* 18, 1487. <https://doi.org/10.3390/ijerph18041487>.
- To, P., Eboreime, E., Agyapong, V.I.O., 2021. The impact of wildfires on mental health: a scoping review. *Behav. Sci.* 11, 126. <https://doi.org/10.3390/bs11090126>.
- US EPA, 2020 [WWW Document]. URL: https://aqs.epa.gov/aqsweb/documents/data_mart_welcome.html. accessed 9.8.20.

- US EPA, 2019. Integrated Science Assessment (ISA) for Particulate Matter (Reports & Assessments). United States Environmental Protection Agency, Research Triangle Park, NC.
- Verma, V., Polidori, A., Schauer, J.J., Shafer, M.M., Cassee, F.R., Sioutas, C., 2009. Physicochemical and toxicological profiles of particulate matter in Los Angeles during the October 2007 southern California wildfires. *Environ. Sci. Technol.* 43, 954–960. <https://doi.org/10.1021/es8021667>.
- Walker, K., Herman, M., 2023. TidyCensus: Load US Census Boundary and Attribute Data as “Tidyverse” and “sf”-Ready Data Frames.
- Wegesser, T.C., Pinkerton, K.E., Last, J.A., 2009. California wildfires of 2008: coarse and fine particulate matter toxicity. *Environ. Health Perspect.* 117, 893–897. <https://doi.org/10.1289/ehp.0800166>.
- Wei, Q., Ji, Y., Gao, H., Yi, W., Pan, R., Cheng, J., He, Y., Tang, C., Liu, X., Song, S., Song, J., Su, H., 2022. Oxidative stress-mediated particulate matter affects the risk of relapse in schizophrenia patients: air purification intervention-based panel study. *Environmental Pollution* 292, 118348. <https://doi.org/10.1016/j.envpol.2021.118348>.
- Westerling, A.L., Hidalgo, H.G., Cayan, D.R., Swetnam, T.W., 2006. Warming and earlier spring increase western U.S. Forest wildfire activity. *Science* 313, 940–943. <https://doi.org/10.1126/science.1128834>.
- Wildfire Smoke: A Guide for Public Health Officials, 2021. <https://www.airnow.gov/v/publications/wildfire-smoke-guide/wildfire-smoke-a-guide-for-public-health-officials/>.
- Yearby, R., Clark, B., Figueroa, J.F., 2022. Structural racism in historical and modern US health care policy. *Health Aff.* 41, 187–194. <https://doi.org/10.1377/hlthaff.2021.01466>.
- Yi, S.S., Kwon, S.C., Suss, R., Doãn, L.N., John, I., Islam, N.S., Trinh-Shevrin, C., 2022. The mutually reinforcing cycle of poor data quality and racialized stereotypes that shapes Asian American health. *Health Aff.* 41, 296–303. <https://doi.org/10.1377/hlthaff.2021.01417>.

Glossary

AMI: acute myocardial infarction
 API: non-Hispanic Asian and Pacific Islander
 AURI: acute upper respiratory infections
 CLRD: chronic lower respiratory disease
 EDVs: emergency department visits
 IHD: ischemic heart disease
 PVD: peripheral vascular disease
 TIA: transient ischemic attack